

# **DAI 1000**

# **Digital and AI Literacy**

**L1 - Course Overview**

# Teaching Team

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# Facets of the Course

- DAI1000 is one of the University Core courses
  - all students must take and pass
- An introductory course on AI and digital technologies
  - with no pre-requisite required
- There will also be some data analysis practices
- Hopefully, DAI1000 will help you excel in your undergraduate

# When to take DAI and how to pass it

Dear Professor Liao,

Hope this email finds you well. My name is [REDACTED] and my student ID is [REDACTED]. I am currently a second-year student in [REDACTED]. I am writing this email to apply for attending your session of [REDACTED].

I am aware that this course is only open to first-year students and could not be registered by sophomores. However, I did not take the course as recommended by the registry because I needed extra credits to attend the [REDACTED] course in the previous academic year.

Since [REDACTED] is a required course for all undergraduates, I hope to have the chance to attend your [REDACTED] course to fulfill the requirements of graduation. I also hope to have the course materials of the first two lectures since students not enrolled in the session do not have the access to course materials on BlackBoard. I

尊敬的廖老师：

您好！

我是来自 [REDACTED]，很抱歉大一上学期最后一节课有事情没来导致了我第一学期 [REDACTED] 成绩不够理想需要重修，因此我特别写这封邮件来请求加课。我保证我这学期绝对不会拖欠作业，保证认真完成小组作业，保证出勤率。请您批准我能来到 [REDACTED] 的课堂学习，谢谢！

尊敬的教授：

您好！很抱歉打扰您，因大一学年ITE课程结课评价因个人私事缺席导致教授遗憾未达到修课合格标准，这学期进行课程安排后决定添加重修 [REDACTED] 您任教的时间段，加课系统上我未满足前提条件故而无法通过正式加课进行选课，现提前向您请示可否通过交表添加您的课程，请您有空时回复这封邮件，诚挚感谢

# Course Details

- It carries one unit
- One lecture per week
- 14 teaching weeks:
  - 12 traditional lectures
  - 2 assessment sessions (W7-Midterm & W14-Final)

# Attendance Policy

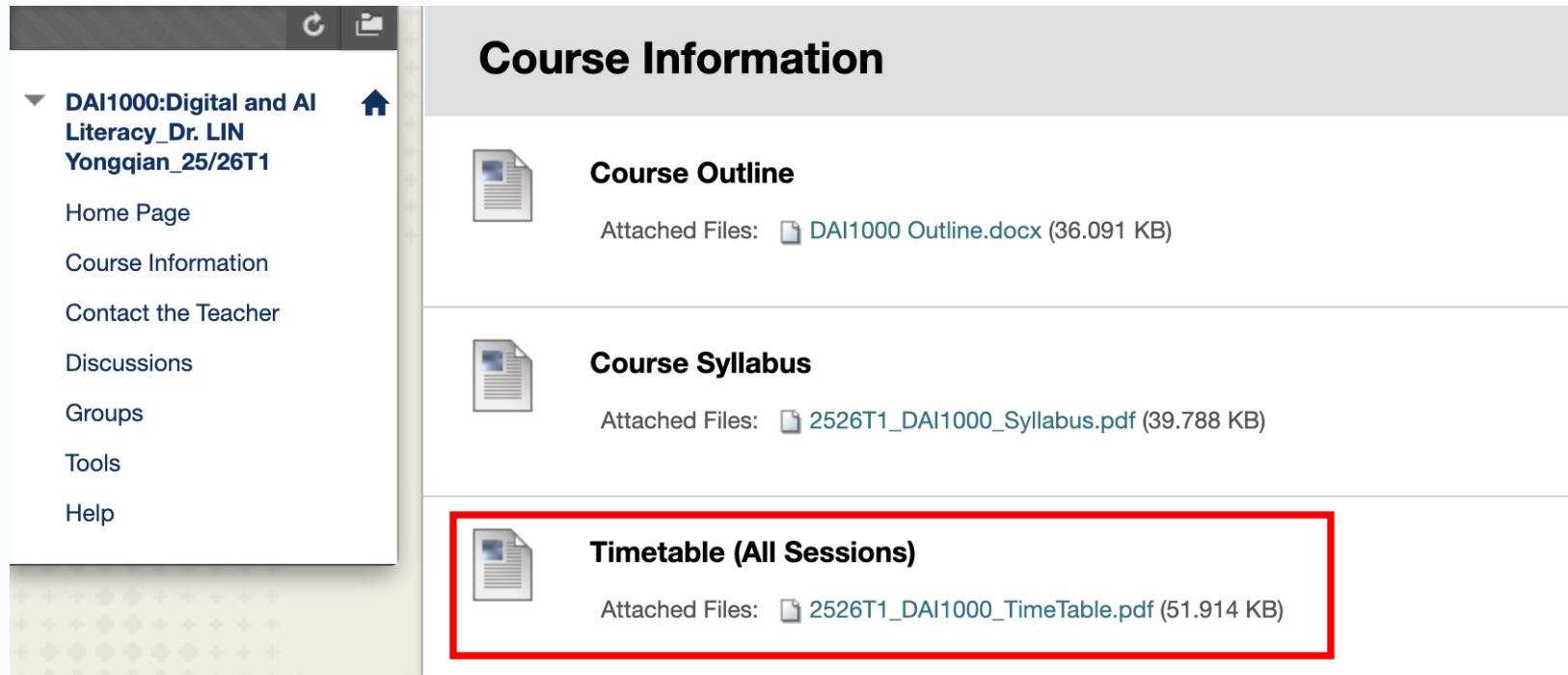
- We will take attendance
- You will be given an F grade if you fail to attend three lessons or more without proper justification
- If you get an F grade, you must repeat the course until you pass

# Attendance Policy

- We do not accept random request for leave, either verbally or by email
- You need inform me about your absence with valid reasons and supporting documents (e.g., doctor letter) **before the class**
- If not, you will be marked as absent!
- It is your responsibility to inform us about your status

# Make-up Classes

- Make-up sessions are possible
- Please email us before attending a make-up class
- Please see the timetable on BB for all sessions



**DAI1000: Digital and AI Literacy\_Dr. LIN Yongqian\_25/26T1**

- Home Page
- Course Information
- Contact the Teacher
- Discussions
- Groups
- Tools
- Help

### Course Information

**Course Outline**  
Attached Files: [DAI1000 Outline.docx](#) (36.091 KB)

**Course Syllabus**  
Attached Files: [2526T1\\_DAI1000\\_Syllabus.pdf](#) (39.788 KB)

**Timetable (All Sessions)**  
Attached Files: [2526T1\\_DAI1000\\_TimeTable.pdf](#) (51.914 KB)

These documents provide the course objectives, course syllabus, and grade descriptor

# Make-up Classes

| TimeTable of DAI1000 |                                                 |                                               |                                                |                                                  |                                                 |
|----------------------|-------------------------------------------------|-----------------------------------------------|------------------------------------------------|--------------------------------------------------|-------------------------------------------------|
|                      | Monday                                          | Tuesday                                       | Wednesday                                      | Thursday                                         | Friday                                          |
| 8:30                 |                                                 | L03<br>TA 201<br>8:30 - 9:20<br>LIAO Ruizhi   | L09<br>TA 202<br>8:30 - 9:20<br>LIN Yongqian   | L13<br>TA 201<br>8:30 - 9:20<br>LIN Yongqian     |                                                 |
| 9:00                 |                                                 |                                               |                                                |                                                  |                                                 |
| 9:30                 |                                                 | L04<br>TA 201<br>9:30 - 10:20<br>LIAO Ruizhi  | L10<br>TA 202<br>9:30 - 10:20<br>LIN Yongqian  | L14<br>TA 201<br>9:30 - 10:20<br>LIN Yongqian    | L06<br>TxB 201<br>9:30 - 10:20<br>LIAO Ruizhi   |
| 10:00                |                                                 |                                               |                                                |                                                  |                                                 |
| 10:30                |                                                 |                                               |                                                |                                                  | L07<br>TxB 201<br>10:30 - 11:20<br>LIAO Ruizhi  |
| 11:00                |                                                 |                                               |                                                |                                                  |                                                 |
| 11:30                |                                                 |                                               |                                                |                                                  | L08<br>TxB 201<br>11:30 - 12:20<br>LIAO Ruizhi  |
| 12:00                |                                                 |                                               |                                                |                                                  |                                                 |
|                      |                                                 |                                               |                                                |                                                  |                                                 |
|                      |                                                 |                                               |                                                |                                                  |                                                 |
| 13:30                | L02<br>ZHIX 110<br>13:30 - 14:20<br>LIAO Ruizhi | L05<br>TC 106<br>13:30 - 14:20<br>LIAO Ruizhi | L11<br>TA 202<br>13:30 - 14:20<br>LIN Yongqian | L01<br>ZHIX 110<br>13:30 - 14:20<br>LIN Yongqian | L15<br>TxB 201<br>13:30 - 14:20<br>LIN Yongqian |
| 14:00                |                                                 |                                               |                                                |                                                  |                                                 |
| 14:30                |                                                 | L12<br>TC 106<br>14:30 - 15:20<br>LIAO Ruizhi |                                                | L16<br>TxB 201<br>14:30 - 15:20<br>LIN Yongqian  |                                                 |
| 15:00                |                                                 |                                               |                                                |                                                  |                                                 |
| 15:30                |                                                 |                                               |                                                |                                                  |                                                 |

## Three Modules

- History, concept, & related technologies
- Core AI techniques and applications
- AI in Society: ethics, impacts, and future

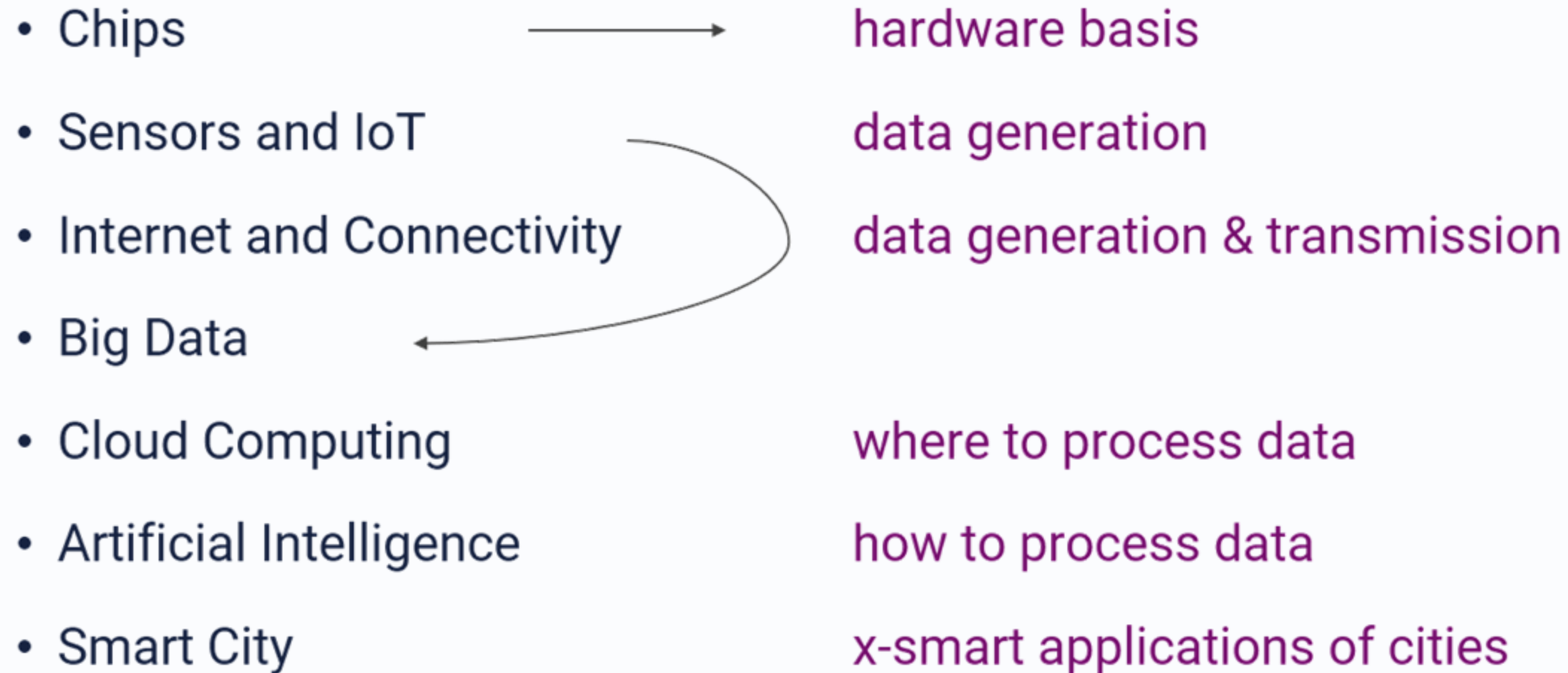
| Week | Lecture Content                         |
|------|-----------------------------------------|
| 1    | Course Overview                         |
| 2    | Information Technology Developments     |
| 3    | AI and Technology Convergence I         |
| 4    | AI and Technology Convergence II        |
| 5    | Machine Learning Essentials I           |
| 6    | Machine Learning Essentials II          |
| 7    | Midterm Assessment                      |
| 8    | Generative AI and Creative Applications |
| 9    | AI Agents and Embodied AI               |
| 10   | AI and Cybersecurity                    |
| 11   | AI Applications for Society             |
| 12   | AI Ethics and Social Impacts            |
| 13   | AI Limitations and Trends               |
| 14   | Final Assessment and Wrap-up            |

# Module 1

- Information Technology Developments
  - IoT, Cloud, big data, and tech evolutions etc
- AI and Technology Convergence I
  - AI history, milestones, and foundational concepts
- AI and Technology Convergence II
  - Evolving technologies that contribute to AI progress

# Module 1

- Information Technology Developments



# Module 1

- AI and Technology Convergence I
  - AI history, milestones, and foundational concepts
- AI and Technology Convergence II
  - Evolving technologies that contribute to AI progress

AI deals with questions of how to see (vision), move (robotics), communicate (natural language, speech), and learn (memory, reasoning, classification).

AI represents a concerted effort to simulate human intelligence in terms of information processes and explore potential ways to automate these processes to obtain desirable outcomes.

# Module 2

- Machine Learning Essentials
  - Different types of machine learning and deep neural networks
- Generative AI and Creative Applications
  - Text, image, video, and copyright issues
- AI Agents and Embodied AI
  - Task automation and AI action in the real world
- AI and Cybersecurity
  - Loopholes, attacks, and defending measures

# Module 3

- AI Applications for Society
  - AI applications in science, business, healthcare, education, etc
- AI Ethics and Social Impacts
  - Fairness, biases, privacy, and social impacts
- AI Limitations and Trends
  - Limitations, pitfalls, and future trends of AI

# Assessments

- **Midterm (45%)** : Data analysis
- **Final quiz (45%)** : Single/multiple-choice questions
- **Participation (10%)**: Attendance will be referred

# Assessments

- Midterm (45%) : Data analysis

Lesson 2 Some Useful Functions

1 Happy day (Y/N)

2 yes

3 no

4 deficit!

5 deficit!

6 yes

7 yes

8 yes

9 yes

10 no

11

12

13 Row:

14 Column:

15 Row1, col1:

16 Row3, col3:

Day

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

Apple sales ( ¥ )

58

47

10

20

77

80

98

90

46

63

13

3

1

-11

Profit ( ¥ )

33

23

-11

-5

57

50

48

30

16

25

589

589

10

98

1

-11

Rent ( ¥ )

5

"VLOOKUP"

finds things by

a key word

5:51 / 8:08

Download video file: [2\\_Useful\\_Functions.mp4](#)

A10

=IF(D10>=30, "Yes",IF(D10<0,"Deficit!", "No"))

|    | A                    | B   | C                 | D            | E          | F | G | H | I | J | K | L | M |
|----|----------------------|-----|-------------------|--------------|------------|---|---|---|---|---|---|---|---|
| 1  | Happy day (Y/N)      | Day | Apple sales ( ¥ ) | Profit ( ¥ ) | Rent ( ¥ ) |   |   |   |   |   |   |   |   |
| 2  | Yes                  | 1   | 58                | 33           | 5          |   |   |   |   |   |   |   |   |
| 3  | No                   | 2   | 47                | 23           |            |   |   |   |   |   |   |   |   |
| 4  | Deficit!             | 3   | 10                | -11          |            |   |   |   |   |   |   |   |   |
| 5  | Deficit!             | 4   | 20                | -5           |            |   |   |   |   |   |   |   |   |
| 6  | Yes                  | 5   | 77                | 57           |            |   |   |   |   |   |   |   |   |
| 7  | Yes                  | 6   | 80                | 50           |            |   |   |   |   |   |   |   |   |
| 8  | Yes                  | 7   | 98                | 48           |            |   |   |   |   |   |   |   |   |
| 9  | Yes                  | 8   | 90                | 30           |            |   |   |   |   |   |   |   |   |
| 10 | No                   | 9   | 46                | 16           |            |   |   |   |   |   |   |   |   |
| 11 | No                   | 10  | 63                | 25           |            |   |   |   |   |   |   |   |   |
| 12 |                      |     |                   |              |            |   |   |   |   |   |   |   |   |
| 13 | Apple sum:           |     | 589               |              |            |   |   |   |   |   |   |   |   |
| 14 | Apple sum(built-in): |     | 589               |              |            |   |   |   |   |   |   |   |   |
| 15 | Minimum:             |     | 10                |              |            |   |   |   |   |   |   |   |   |
| 16 | Maximum:             |     | 98                |              |            |   |   |   |   |   |   |   |   |
| 17 | Row1, col1:          |     | 1                 |              |            |   |   |   |   |   |   |   |   |
| 18 | Row3, col3:          |     | -11               |              |            |   |   |   |   |   |   |   |   |
| 19 |                      |     |                   |              |            |   |   |   |   |   |   |   |   |
| 20 |                      |     |                   |              |            |   |   |   |   |   |   |   |   |
| 21 |                      |     |                   |              |            |   |   |   |   |   |   |   |   |
| 22 |                      |     |                   |              |            |   |   |   |   |   |   |   |   |
| 23 |                      |     |                   |              |            |   |   |   |   |   |   |   |   |
| 24 |                      |     |                   |              |            |   |   |   |   |   |   |   |   |

"VLOOKUP"

finds things by

a key word

# Assessments

- **Final quiz (45%)** : Single/multiple-choice questions

## QUESTION 12

1 points

**A cyber attack in which a swarm of bots controlled by a malicious server attacks a legitimate server to overwhelm its service is called:**

- ☐ A spoofing attack
- ☐ A brute force attack
- ☐ A privilege escalation attack
- ☐ A denial of service attack

## QUESTION 17

1 points

**A student would like to use a picture for her IT video project. How can she do it?**

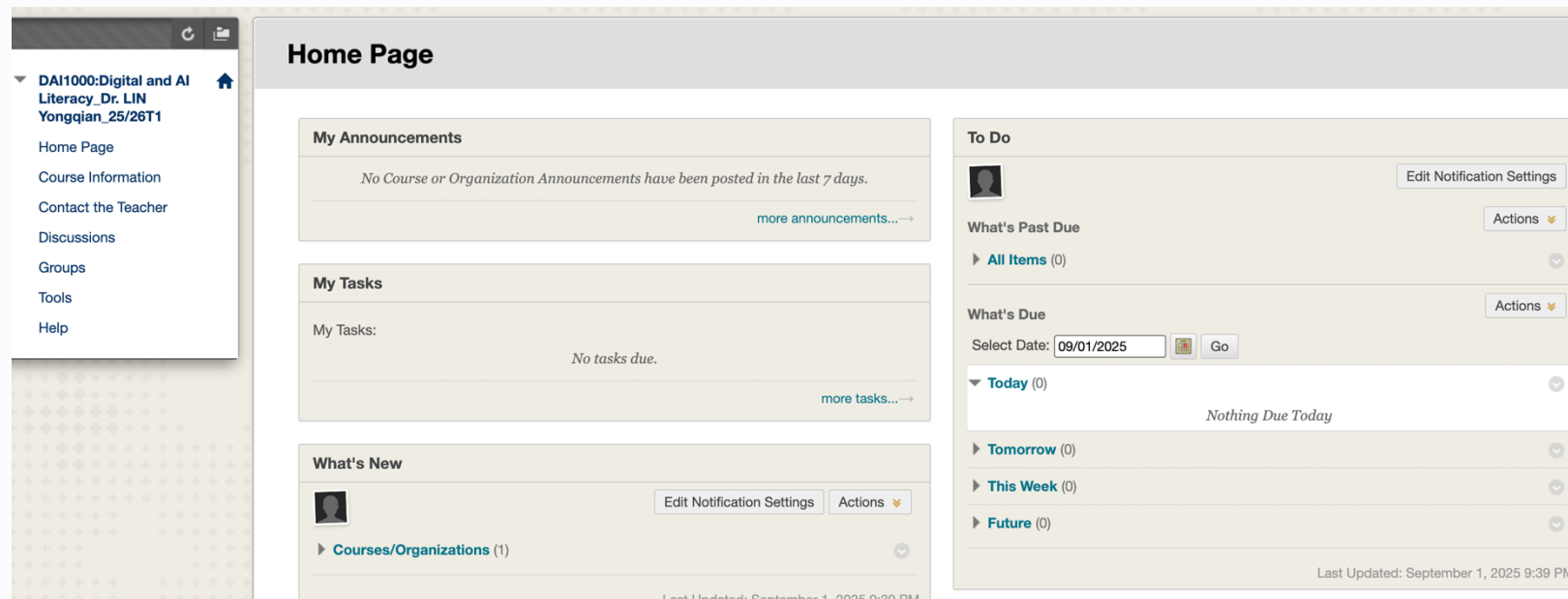
- i. Draw it by herself using Photoshop
- ii. Search on the Internet and copy the picture into her video
- iii. Take a photo of a picture in a magazine and remove the watermark using Photoshop
- iv. Provide a proper citation to the source of the picture at the end of her video

| Grade                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DISTINCTION</b>      | Exceptional performance in all aspects of the course. Demonstrates a deep understanding of how IT and AI technologies transform lifestyles and impact societal development. Shows a strong ability to acquire new knowledge and skills independently. Proficient in using the Internet with excellent etiquette and capable of implementing basic and advanced cybersecurity measures while staying vigilant against threats. Exhibits solid knowledge of information management, ethical principles, and local and international legal and cultural standards. Displays outstanding communication skills and a commitment to leveraging IT and AI responsibly and effectively. |
| <b>PASS</b>             | Good performance in all aspects of the course. Demonstrates a satisfactory understanding of how IT and AI technologies influence lifestyles and societal development. Shows the ability to acquire new knowledge and skills independently. Proficient in using the Internet with acceptable etiquette and capable of implementing basic cybersecurity measures. Possesses adequate knowledge of information management, ethical standards, and local and international legal and cultural norms. Displays good communication skills and a cooperative teamwork spirit.                                                                                                          |
| <b>FAIL &lt; 50/100</b> | Performance does not meet the minimum requirements of the course's assessment tasks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

If you miss one test, you may get FAIL.

# Online Resources and Consultation

- Course Materials and e-learning: <https://bb.cuhk.edu.cn>



- Need help with this course? —Contact your lecturer via email

# Expectations

- Independent
- Teamwork
  - Critical in the University and workplace
- Punctuality
  - For classes and coursework
- Always check your CUHK email
- Honesty
  - Avoid plagiarism and cheating in quizzes and tests

# Honesty in Academic Work

- CUHKSZ places very high importance on honesty in academic work submitted by students, and adopts a policy of **zero tolerance** for cheating and plagiarism
- Any related offence will lead to disciplinary actions, including **termination of studies** at the University

# Questions?